



S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded
Department of Production Engineering
In-Semester Examination - II (Sem-I) Academic Year: 2025-26

Class: Final Year B. Tech	Subject: Advanced Manufacturing Processes	Subject Code: PEC-PE-405
Date: 15/10/2025	Time: 14:30 TO 15:15	Max. Marks: 20

Note

- Assume suitable data whenever necessary and state it clearly.
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.

Q. N.		M	CO	BT
Q.1-A	Explain squeeze casting in detail.	10	CO2	BT1
	OR			
Q.1-B	Describe laser beam welding machine.	10	CO3	BT2
Q.2-A	Explain continuous casting in detail.	10	CO2	BT1
	OR			
Q.2-B	Discuss ultrasonic welding in detail.	10	CO3	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO1. Understand the advanced machining processes.

CO2. Suggest the suitable advanced casting processes.

CO3. Select proper methodology for advanced welding processes.

CO4. Study different advanced metal forming processes.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded
Department of Production Engineering
Academic Year: 2025-26 (semester I)
In-Semester Examination / Evaluation - II

Class: Final Year B.Tech.

**Course Name: Operations and
Materials Management**

Course Code: PCC-PE401

Date: 14.10.2025

Time: 14:30 to 15:15

Max. Marks: 20

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Q.N.	Solve the following questions (Any Two)	Marks	CO	BT
Q.1	a Define 'Material Requirement Planning' and describe the various inputs to MRP system.	05	CO2	BT1
	b What is meant by 'Inventory Control'? Explain any four techniques used for effective inventory management.	05	CO2	BT1
Q.2	a An oil engine manufacturer purchases lubricants at the rate of Rs. 42 per piece from a vendor. The requirements of these lubricants are 1800 per year. What should be the ordering quantity per order if the cost per placement of an order is Rs. 16 and inventory carrying charges per rupee per year is 20 paise.	04	CO2	BT4
	b What is meant by Enterprise Resource Planning? Explain the three major reasons why companies undertake ERP?	06	CO2	BT2
Q.3	a A company need 600 units per month, the procurement cost is Rs. 36 per order, the cost of holding it in stock is Rs. 1.20 per unit per year, determine the quantity that should be procured at a time to optimise the cost involved. If the consumption of the above item increases to 40 numbers per day and its actual inventory cost is Rs. 0.50 per unit per month, what will be the revised EOQ?	06	CO2	BT5
	b Explain the scope of materials planning and control.	04	CO2	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Recognize operation strategy concept.
CO 2	Develop the material management planning for final product.
CO 3	Analyze the different push and pull production system tools.
CO 4	Implement the concept of TOC and MOST techniques.
CO 5	Understand SCM used to improve the Supply Chain performance



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Production Engineering
In-Semester Examination / Evaluation II
Academic Year: 2025-26 (semester I)

Class: Final Year B.Tech.

Elective: Project Management

Course Code: PEC-PE407

Date: 09.10.2025

Time: 14:30 to 15:15

Max. Marks: 20

- Attempt any two questions
- Assume suitable data whenever necessary and state it clearly.
- Figures to right indicate full marks.

Q.N.						Marks	CO	BT
Q.1	Consider a project with the following details. Draw the CPM network (AOA) and determine the earliest and latest times by forward pass and backward pass method. Determine critical path and the project completion time.					10	CO4	BT2
		Activity	Start Node	End Node	Activity Time			
		A	1	2	11			
		B	1	3	3			
		C	1	4	10			
		D	4	5	3			
		E	3	5	1			
		F	2	6	2			
		G	5	6	1			
		H	6	7	1			
		I	6	8	1			
		J	7	8	5			
Q.2	Suppose that a project required an initial cash investment of \$24,000 and was expected to generate inflows of \$10,000, \$10,000, and \$10,000 for the next three years. Further, assume that our company's required rate of return for new projects is 11%. Determine IRR. Is this project worth funding?					10	CO1	BT4
Q.3	A project might have the following NPVs at the following discount rates (Discount rate in %, NPV in thousand Pounds). Plot the data on a graph paper and determine IRR.					10	CO2	BT3
		Discount rate	5	10	15	20		
		NPV	5300	700	-1500	-3200		

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	Explore the feasibility of a project by applying financial and environmental criteria.
CO2	Learn various stages of life cycle and its implementation.
CO4	Understand components of the critical path and they can utilize PERT analysis to plan, manage and evaluate a large project.



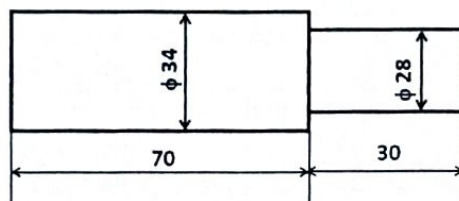
Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering
In-Semester Examination / Evaluation II
Academic Year: 2025-26 (Semester I)

Class: Final Year B.Tech.	Course: Costing and Estimation	Course Code: PEC-PE406
Date: 09.10.2025	Time: 02.30 pm to 03.15 pm	Max. Marks: 20

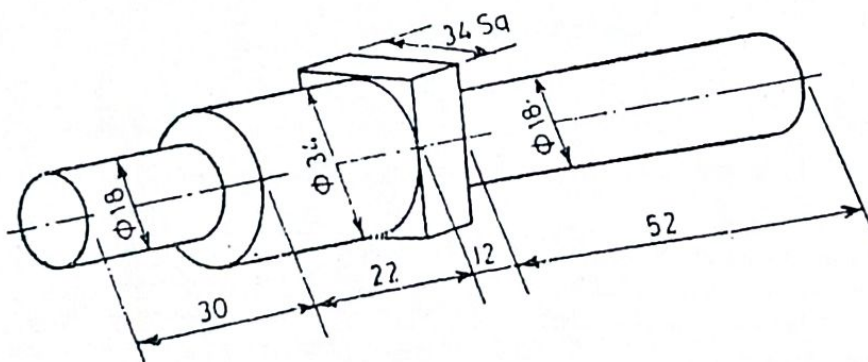
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Q.N.		Marks	CO	BT
Q.1	Find the time required to turn 40 mm diameter bar to the dimensions shown in the following Fig. The cutting speed is 15 m/min, and the feed is 1.0 mm per revolution. All cuts are 3 mm deep.	10	CO4	BT3



Q.2	Estimate the cost of the material required to manufacture 100 pieces of copper conductor as shown below. The density of the copper material is 8000 kg/m ³ . Consider that 10% of the material is lost in the forging and finishing process. The market price of the raw mild steel material is Rs. 1000 per kg. All dimensions are in mm.	10	CO3	BT4
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=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

- On successful completion of this course, students should be able to:
- Learn the basic concepts of cost, estimation, and depreciation fund calculation.
 - Compute costs for various manufacturing processes like forging, welding, foundry etc.
 - Understand the process of job costing, activity-based costing, cost accounting and budgetary control.
 - Interpret during decision making, the concepts of CVP analysis, cost control techniques including time value of money.
 - Implement the basics of engineering economics and financial management for profit making by the organization.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering.
In-Semester Examination / Evaluation II
Academic Year: 2025-26 (semester I)

Class: Final Year B.Tech.

Course: CAD/CAM & Robotics. Course Code: PEC-PE404

Date: 10.10.2025

Time: 14:30 to 15:15

Max. Marks: 20

Note:

Reg.No:

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Q.N.			Mark s	CO	BT
Q.1		Answer the following, (Any 2)			
	a	Explain the concept of surface modelling along with its advantages and disadvantages.	06	CO2	BT2
	b	Write a short note on the data exchange formats used in CAD/CAM software.	06	CO1	BT1
	c	What is NC? Explain the steps involved in NC manufacturing (NC procedure), and state its advantages and limitations.	06	CO3	BT1,2
Q.2	a	Discuss the boundary representation (B-rep).	04	CO2	BT3
	b	Explain the different features used in FBM.	04	CO2	BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	Learn CAD hardware and software for variety of applications
CO2	Compare different alternative facilities in CAD software
CO3	Implement the NC part programming and its application
CO4	Understand the different concepts in manufacturing system and its application for industrial automation.
CO5	Describe the robot and its different industrial applications.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering
Academic Year: 2025-26 (semester I)
In-Semester Examination / Evaluation II

Class: Final Year B.Tech. Course Name: Tools for Six Sigma Course Code: PCC-PE403
Date: 11.10.2025 Time: 14:30 to 15:15 Max. Marks: 20

Note:

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- The symbols carry usual meaning.
- Question 1 is compulsory and solve any three questions in Question 2.**

Q.N.		Marks	CO	BT
Q.1	The company manufactures 10,000 pens and, after inspection, identifies 120 defects in total across the various opportunities viz., Ink Quality, Ballpoint, Cap, and Barrel. Find the Defects per Million Opportunities.	02	CO4	BT5
Q.2	A Imagine a manufacturing plant that produces metal bolts. The bolts must have a diameter of 10mm, but slight variations can occur due to machine imperfections. After analyzing the production for a day, the quality assurance team finds: - Upper Specification Limit = 10.05mm - Lower Specification Limit = 9.95mm - Process Mean = 10.01mm - Standard Deviation (from a sample) = 0.02mm Find the Process's Process Capability and Process Capability Index.	06	CO4	BT5
	B Explain SIPOC diagram.	06	CO3	BT3
	C What are the check points for completion of analyse phase?	06	CO3	BT3
	D What are the key steps involved in hypothesis testing?	06	CO4	BT4

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Plan/Identify, BT4 - Motive, BT5 - Measure/Evaluate,

BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand History, Concepts, Definitions, Levels of Six Sigma, Characteristics, and objectives of Six Sigma.
CO 2	Learn Road map for six sigma, DMAIC methodology.
CO 3	Use graphical tools, process mapping, FMEA, Ishikawa diagram.
CO 4	Solve numerical based on CI, ANOVA, HT, Correlation and Regression, cost of poor quality, identify prioritize and select six sigma projects.
CO 5	Solve numerical for parameter optimization using concepts of DOE, Robust design, Response plots, O.A., S/N ratios.
CO 6	Design a product using concept of QFD, and House of Quality.



SGGS INSTITUTE OF ENGINEERING & TECHNOLOGY, VISHNUPURI, NANDED – 431 606

Department of Production Engineering

Academic Year 2024-2025

ISE-2 Examination

Class: B.Tech. (Prod)

DE: 13 /10/25
14.30-15.15

Subject: Productivity improvement techniques (PCC-PE-402)

Max Marks: 20

*CO: Course Outcomes.

NOTE: SOLVE ANY TWO QUESTIONS FROM THE FOLLOWING.

Q.1	An operator has to attend two workstations. The actual time consumption by both the workstations is given below, prepare multiple activity chart on a common time scale and find % utilization of operator and two work-stations, State your conclusions.			10	CO6	BT4
	Sr no	Activity	Time in min(WS-1)	Time in min(WS-2)		
	1	Stopping the machine(unloading)	2	2		
	2	Processing time	25	12		
	3	Loading time	2	5		
	4	Inspection time	3	2		
Q.2	An aluminium disc of 50mm diameter is to be manufactured from a ready stock of 400mmX200mm of available stock. The sheet is 4mm thick. The process operation is to be carried out on a 500 ton press. The process includes major operations such as cutting into strips, blanking, embossing, piercing, degreasing etc. The material is to be issued from stores. The distances of machines from stores are cutting/shearing machine- 1m, Press-12m from cutting machine, embossing press – 6m from press, cleaning-6m from embossing machine, degreasing- next to cleaning, inspection bench -10m from cleaning and stores-12m from inspection bench. Draw flow process chart material type in format for above process, critically examine the same, suggest improvements if any. Only major operations are given minute details and time required have to be assumed. (use any other data if required)			10	CO5	BT3
Q.3	Explain in detail the construction procedure and significance of a travel chart by assuming suitable example and data if required.			10	CO2	BT2